S_24_31

(c) **(i)** Complete the Karnaugh map (K-map) for the Boolean expression:

$\overline{A}.\overline{B}.\overline{C} + \overline{A}.\overline{B}.C + A.\overline{B}.\overline{C} + A.\overline{B}.C + A.B.\overline{C} + A.B.C$

A \ BC	00	01	11	10
0				
1				

[2]

(ii) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(iii) Write the Boolean expression from your answer to part **(c)(ii)** as a simplified sum-of-products.

.....

..... [1]

S_24_32

(c) **(i)** Complete the Karnaugh map (K-map) for this Boolean expression:

$\overline{A}.\overline{B}.\overline{C} + \overline{A}.B.\overline{C} + \overline{A}.B.C + A.\overline{B}.\overline{C} + A.B.\overline{C} + A.B.C$

A \ BC	00	01	11	10
0				
1				

[2]

(ii) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(iii) Write the Boolean expression from your answer to part **c(ii)** as a simplified sum-of-products.

.....

..... [1]

S_24_33

(c) **(i)** Complete the Karnaugh map (K-map) for the Boolean expression:

$\overline{A}.\overline{B}.\overline{C} + \overline{A}.\overline{B}.C + A.\overline{B}.\overline{C} + A.\overline{B}.C + A.B.\overline{C} + A.B.C$

A \ BC	00	01	11	10
0				
1				

[2]

(ii) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(iii) Write the Boolean expression from your answer to part **(c)(ii)** as a simplified sum-of-products.

.....

..... [1]

S_23_33

7 **(a)** Complete the Karnaugh map (K-map) for the following Boolean expression.

$$Z = \overline{A}.\overline{B}.\overline{C}.\overline{D} + \overline{A}.\overline{B}.\overline{C}.D + \overline{A}.B.\overline{C}.\overline{D} + \overline{A}.B.\overline{C}.D + A.B.\overline{C}.\overline{D} + A.B.\overline{C}.D$$

CD \ AB	00	01	11	10
00				
01				
11				
10				

[2]

(b) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(c) Write the Boolean logic expression from your answer to **part (b)** as a simplified sum-of-products.

Z = ...

..... [2]

(d) Use Boolean algebra to give your answer to **part (c)** in its simplest form.

Z = ... [1]